Ideation Phase
Define the Problem Statements

Date	13 February 2026
Team ID	LTVIP2026TMIDS79181
Project Name	Smart Sorting – Rotten Fruit Detection using Machine Learning
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create clear problem statements from the customer's point of view to guide solution design. These statements help the team empathize with users and focus on real-world challenges faced by stakeholders in digital payment systems.

Problem Statements (PS)

PS-1 (Fruit Vendor / Shop Owner Persona)

I am a fruit vendor who sells fruits to customers daily in the market.

I'm trying to provide fresh and good-quality fruits to my customers.

But it is difficult to manually identify and separate rotten fruits quickly.

Because manual sorting takes time and sometimes rotten fruits are missed during inspection.

Which makes me feel worried about customer dissatisfaction, food waste, and financial losses.

PS-2 (Warehouse / Distributor Persona)

I am responsible for sorting and distributing large quantities of fruits in warehouses or supply chains.

I'm trying to efficiently separate fresh fruits from rotten ones before distribution.

But manual inspection is slow and inconsistent when dealing with large volumes of fruits.

Because human inspection may miss defects or spoilage due to fatigue or time pressure.

Which makes me feel concerned about product quality, operational efficiency, and increased waste.

PS-3 (Food Quality Inspector / Agricultural Industry Persona)

I am responsible for maintaining quality standards in fruit supply chains.

I’m trying to ensure that only healthy and fresh fruits reach consumers.

But current methods rely heavily on manual checking and lack automated systems.

Because traditional methods cannot quickly analyze fruit quality at scale.

Which makes me feel concerned about food safety, quality control, and reducing post-harvest losses.

Proposed Direction (Solution Focus)

Develop a **Machine Learning–based Smart Sorting System** that analyzes fruit images and automatically detects whether a fruit is fresh or rotten. The system will use image processing and deep learning techniques to classify fruits accurately. This solution will help vendors, distributors, and quality inspectors quickly identify spoiled fruits, reduce food waste, improve efficiency, and ensure better quality in the fruit supply chain.